Object Segmentation Based on Pseudo Supervision Relearning Under Extremely Weak Annotations for Remote Sensing Images

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***Abstract*— Weakly annotated object segmentation for remote sensing images (RSIs) has attracted lots of attention due to its low labeling costs. However, annotating a huge amount of multiclass RSIs with image labels is still highly dependent on specific expert knowledge and, thus, requires considerable labeling costs. In this article, intending to further alleviate the labor-intensive labeling costs, we introduce a novel extremely weak annotation condition. In this condition, a substantial portion of samples are unlabeled, while merely a small number of samples are labeled with inexact image-level annotations. To achieve object segmentation under extremely weak annotations, we propose pseudo supervision relearning (PSRL), a novel three-stage framework with the core insight of effectively harnessing the potentially valuable supervision clues stored in abundant unlabeled data. In the first stage, the extremely weak annotations are switched to fine-grained but noisy pseudo supervision with the aid of image-level semantic learning and attention-guided data augmentation. Then, a category-aware dataset resplit strategy based on the masking perturbation mechanism is designed, aiming at adaptively selecting high-quality pixelwise pseudomasks from the artificially generated pseudo supervision and achieving class-balanced reliable–unreliable labels division. Ultimately, we devise a novel dynamic thresholding strategy (DTS)-guided relearning network to take full advantage of the valuable semantic information in the resplit dataset. Experimental results on two public RSI datasets show the effectiveness of the proposed framework. Utilizing less supervised information, the proposed method yields competing results compared to weakly supervised learning-based methods with complete image-level annotations.**

***Index Terms*— Object detection, pseudo supervision relearning (PSRL), remote sensing image (RSI) processing, semantic segmentation, weakly supervised learning.**

I. Introduction

OBJECT segmentation for remote sensing images (RSIs) is a new technique arising with the development of remote sensing technology [1], [2], which can provide accurate semantic and spatial position information for geographic objects of interest. Thanks to the vigorous development of deep learning, many advanced deep networks have successively emerged and greatly promoted the intellectual interpretation of RSIs, such as SpectralGPT [3] built by transformer, the first foundation model specially designed for remote sensing, which can provide highly reliable information for downstream tasks such as object segmentation. Besides, some models based on traditional convolutional neural networks [4] have also achieved excellent success, which promotes the popularity of object segmentation tasks in various applications such as urban planning [5], [6], environment monitoring [7], [8], [9], resource exploration [10], [11], and military detecting [12], [13], [14]. Recently, due to its cost-effective labeling process, weakly supervised object segmentation (WSOS) has gained popularity and achieved tremendous progress [15], [16], [17]. Among them, the image-level WSOS methods have gained the most interest since they merely require the class labels of semantic objects existing in the image as supervision information.

However, for large amounts of multiclass RSIs, acquiring image-level weak annotations still requires significant labeling costs. In addition, due to factors of data perturbation and annotation demand for domain knowledge, there is an issue regarding the scarcity of annotated data, especially in some special remote sensing scenarios. In the meantime, there exists a large amount of unlabeled data that is easier to obtain [18], [19], with huge potential application value maintained.

To further reduce the labeling costs, different from the existing forms of weak annotations that all training samples are labeled with inexact annotations such as image-level labels, we establish a novel annotation condition namely **extremely weak annotation** for the first time, which can be regarded as a weaker version of conventional weak annotation and is suitable for the limited conditions of both labeling quantity and labeling precision. Specifically, the extremely weak annotation is characterized by both inexactness and incompleteness, which is reflected in the following two aspects: 1) the accessible annotations are inexact and coarse-grained, i.e., only at the image level and 2) merely a small portion of training samples are labeled, while most of the samples are unlabeled [as shown in Fig. 1(c)]. Hence, it relies on much slighter human labeling than conventional weak annotation.

However, the absence of image-level category labels for most of the training samples (e.g., merely ten samples labeled with image-level tags per category in our experiment) in extremely weak annotations will inevitably lead to the uncertainty and incompleteness of supervision signals.

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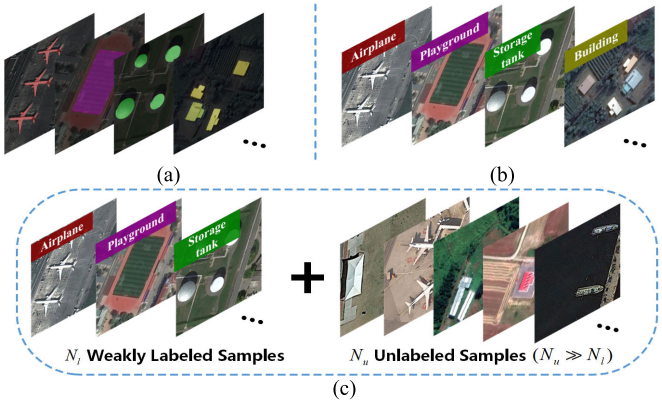


Fig. 1. Illustration of training data labeled with different annotation conditions. (a) and (b) Full and weak annotations, respectively, in which all of the training samples are annotated with pixel-level labels and image-level labels, respectively. (c) Extremely weak annotations, in which merely a small number of samples are labeled with image-level weak annotations, while most of the rest are unlabeled.

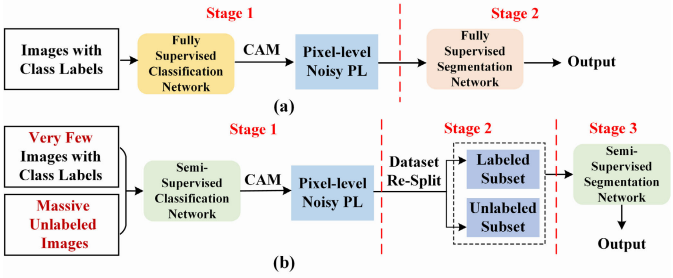


Fig. 2. Training paradigms of (a) conventional WSOS framework and (b) our object segmentation framework under the condition of extremely weak annotation based on the PSRL mechanism.

In addition, taking into consideration the unique characteristics of RSIs including diverse ground object information and complicated background interferences, learning to transform limited image-level object categories into pixelwise semantic predictions is particularly challenging. Consequently, the key to learning geographic object segmentation under extremely weak annotations is to fully excavate and exploit the valuable knowledge from massive unlabeled samples and utilize the extracted features to assist the stream of labeled samples.

As shown in Fig. 2(a), the conventional WSOS framework mainly consists of two cascaded stages, i.e., first generating pseudolabels (PLs) from the classification network as object seeds and then employing the generated seeds as pseudo supervision for pixel-level semantic learning and inference [20], [21]. According to the above analysis, the core challenge for the novel extremely weak annotation condition is distinct from the existing weakly supervised learning-based pipeline. However, at present, there are few studies about object segmentation under the condition of extremely weak annotations [22], [23]. Among them, Li et al. [23] exploit unsupervised representation learning techniques to acquire valuable semantic representation in the unlabeled samples. Nevertheless, the response activation clustering technique used to learn image features in [23] relies on high-dimensional clustering computation, which will lead to huge computational costs and further decrease the training efficiency in complex remote sensing scenarios. Therefore, the method in [23] is difficult to directly apply in remote sensing scenarios, and it is necessary to develop an extremely weak annotated object segmentation method for RSIs.

In this article, we start from the perspective of innovating the learning mechanism and propose a novel three-stage pseudo supervision relearning (PSRL) framework. In order to fully exploit the potential semantic information in massive unlabeled samples, we specially design an adaptive dataset resplit strategy and a dynamic thresholding-guided relearning mechanism to enhance model performance. Specifically, as shown in Fig. 2(b), PSRL decouples the extremely weak annotated object segmentation task into three phases to achieve a progressive transition of the supervision from insufficient and inexact annotations to abundant and precise supervision. In the first stage, image-level semantic learning is achieved based on semisupervised classification training, and an attention-guided augmentation strategy is designed to yield more accurate image-level category annotations for RSIs featuring randomly located geographic objects. Then, a category-aware masking perturbation mechanism is proposed to resplit the datasets according to the image-level similarity and confidence values, reducing the negative effect caused by the lack of true category labels. Ultimately, a dynamic thresholding-guided relearning network is trained with the resplit dataset, serving as a PSRL phase to further excavate the semantic clues contained in the pseudo supervision while also avoiding confirmation bias. Overall, the above three phases form the PSRL framework, which can take advantage of the original unlabeled data as much as possible.

To the best of our knowledge, PSRL is the first object segmentation method for RSIs under the condition of extremely weak annotations. The main contributions are given as follows.

1) *PSRL:* We propose a PSRL framework under the challenging condition of extremely weak annotations to fully excavate knowledge from massive unlabeled samples and limited labeled samples with image tags and achieve the progressive improvement of the model segmentation performance.
2) *Category-Aware Dataset Resplit:* To adaptively select high-quality PLs from the generated noisy labels, a category-aware dataset resplit strategy based on the masking perturbation mechanism is designed for class-balanced reliable–unreliable pseudomasks split, which helps to relieve the adverse impact caused by the lack of abundant real category labels.
3) *Dynamic Thresholding-Guided Relearning:* A novel dynamic thresholding strategy (DTS) is proposed to adaptively adjust the confidence threshold for relearning from the resplit dataset, which selects the samples in an easy-to-hard way and, thus, significantly helps alleviate confirmation bias caused by the noise interferences contained in self-generated artificial labels.

In addition, experiments on two public RSI datasets show that the proposed method achieves satisfactory performance under the condition of extremely weak annotations. Meanwhile, competitive results are acquired compared to the

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state-of-the-art (SOTA) methods that utilize complete image-level annotations as well.

The remaining content of this article is structured as follows. Section II introduces a brief review of related works about WSOS and semisupervised object segmentation (SSOS). Section III provides the problem definition of extremely weak annotated object segmentation and then introduces the details of the proposed PSRL. Section IV presents the experimental results and overall analysis. Finally, Section V concludes this article.

II. Related Work

In this section, we review the object segmentation methods related to our work under weakly supervised and semisupervised conditions, respectively.

A. Weakly Supervised Object Segmentation

WSOS has attracted lots of attention and has been broadly studied recently because of its value in minimizing the labeling cost caused by pixel-level annotations required in fully supervised segmentation, which mainly uses image-level labels [24], [25], [26], bounding-box [27], [28], and scribble labels [29], [30] as supervision signals. Among the multiple types of weak supervision, image-level annotations have become the most prevalent on account of their lowest manual labeling costs. Nevertheless, the lack of concrete clues about target object positions and shapes greatly increases the difficulty of the pixelwise segmentation task [31].

The existing WSOS approaches are mainly based on the class activation mapping (CAM) algorithm [32] to generate seed regions as coarse pseudo supervision, which is employed to train a subsequent segmentation network. Many studies focus on improving the quality of pseudopixel supervision. For example, Wei et al. [24] propose an adversarial erasing algorithm iteratively. By masking the most discriminative region detected in the current iteration round, it can explicitly drive the classification model to explore new target regions in the next iteration round, thus generating more integrated object regions. To achieve significant improvement in the quality of the pseudomask, Araslanov and Roth [25] propose a pseudomask correction based on affinity learning, which uses the random walk algorithm to propagate the initial target region to the pixel region with high semantic similarity in the neighborhood. In addition, some researchers also focus on reducing the influence of noise from PLs by designing denoising loss or label correction strategies [28], [33], [34].

B. Semisupervised Object Segmentation

SSOS based on the semisupervised learning (SSL) mechanism aims at effectively employing massive unlabeled data to improve learning performance with limited manually labeled data and reduce the need for large-scale data annotation. Widely used SSL methods can be mainly divided into two categories: consistency regularization [35], [36], [37], [38] and self-training methods [39], [40].

In detail, the self-training methods generate PLs for unlabeled samples and combine them with labeled data to join in the retraining process [41], [42], which gradually improves the model performance iteratively. The core insight of self-training methods is to effectively utilize the information of unlabeled data to enhance the model's generalization ability. Yang et al. [41] propose a selective retraining mechanism by training unlabeled samples in an easy-to-hard manner, which helps to mitigate the negative effect caused by inaccurate PLs. Ke et al. [42] propose a three-stage self-training framework that enhances the certainty of pseudomasks in a multitask fashion. However, the self-training methods have certain risks when using unlabeled data, which may introduce false labels and cause model performance degradation.

As for the consistency regularization-based methods, the model is encouraged to produce consistent predictions on images with different perturbations [43], [44], [45], which helps to boost the model's generalization ability and performance. The general process is that a weak augmentation is utilized to yield PLs for unlabeled data, and a strong augmentation is used for consistency training. A consistency loss function is introduced to implement consistency regularization to force the model's predictions to be consistent. To effectively propagate supervision information from the limited annotated images to the unlabeled ones, Xu et al. [43] propose a novel prototype-based consistency loss to regularize the distribution of intraclass feature representation.

The prevalent SSL methods mostly utilize a fixed threshold value or manually designed threshold adjusting methods to select reliable PLs for supervising the unlabeled samples. For example, UniMatch [46] maintains a fixed high threshold for selecting highly confident PLs. However, the model performance is highly dependent on the selection of the threshold value since an improper threshold will influence the quality of PLs and lead to a suboptimal model performance.

In this article, to better leverage the unlabeled data, we introduce SSL as an efficient bridge to realize the semantic transition from extremely weak annotations to strong annotations. It is mentioned that in contrast to vanilla SSOS methods, we design a dynamically weighted thresholding scheme to adaptively adjust the threshold values to make better use of pseudo supervision.

III. Methodology

A. Problem Definition

We first establish the proposed condition of extremely weak annotations. Let $\mathbb{D}$ denote the extremely weakly annotated dataset, which is composed of a weakly labeled subset $\mathbb{D}^l$ and an unlabeled subset $\mathbb{D}^u$. Among the dataset, $\mathbb{D}^l = \{(\mathbf{I}_l, \mathbf{Y}_l)\}_{l=1}^{N_l}$ contains N_l labeled samples with image-level weak annotations. The label can be denoted as $\mathbf{Y}_l \in \{0, 1\}^C$, where C represents the total amount of geographic object categories in the training set. $\mathbb{D}^u = \{\mathbf{I}_u\}_{u=1}^{N_u}$ includes N_u unlabeled samples, and $N_u \gg N_l$. In our experiment, each semantic category in the training set contains merely ten image-level weakly labeled samples, and the rest are all unlabeled samples. In addition, both weakly labeled and unlabeled samples in the extremely weak supervised dataset were collected from the same remote sensing platform under

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the same collection conditions. In this article, by fully utilizing the abundant unlabeled RSI samples that are easier to acquire, the extremely weak annotated object segmentation task aims at further reducing the burden of labeling costs while ensuring segmentation accuracy.

B. Overview of the Proposed Method

Given the training dataset $\mathbb{D}$ under extremely weak supervision, in the first stage, a semisupervised classification network $\mathcal{F}_{\mathrm{CLS}}$ is primarily trained to generate image-level pseudolabels of unlabeled samples, and subsequently pixel-level pseudomasks of all samples in the dataset are produced by the CAM algorithm. Through the first stage described above, a new dataset $\mathbb{D}_N$ including images and corresponding pixel-level pseudomasks can be obtained. Considering the absence of a large number of real image category labels under the condition of extremely weak annotation, the generated pseudomasks will unavoidably produce inaccurate judgment results. To tackle the issue of noisy labels, in the second stage, the dataset $\mathbb{D}_N$ is redivided into a labeled subset and an unlabeled subset using the category-aware dataset resplit strategy by employing the quality of pseudomasks as splitting metrics. Ultimately, in the third stage, a relearning network $\mathcal{F}_{\mathrm{SEG}}$ is trained by the DTS, and valuable semantic information existing in the unlabeled samples is further mined to improve the accuracy of the object segmentation results.

The overall framework of the proposed three-stage algorithm is illustrated in Fig. 3, and the details of each stage in the proposed framework are given in the following.

C. Pseudomask Generation Based on Image-Level Semantic Learning

To leverage the massive unlabeled RSI samples under the condition of extremely weak annotations, SSL is intuitively introduced since it can serve as a bridge to shrink the gap between unlabeled and labeled data. Technically, a classification network based on SSL is trained to learn image-level object semantics and assign pseudocategory labels for each unlabeled sample, and the pixel-level pseudomasks can be inferred through postprocessing operations. In particular, an attention-guided data augmentation strategy is designed to generate novel image views from unlabeled RSI samples, aiming at promoting semisupervised classification performance. In this section, we first introduce the baseline architecture of the semisupervised classification network. Then, we provide a detailed explanation of the attention-guided data augmentation strategy.

1) Semisupervised Classification Baseline Network Based on Weak-to-Strong Consistency Regularization: The key idea behind consistency regularization is to apply various data augmentations to encourage consistent predictions for unlabeled samples. Following the typical semisupervised classification method FixMatch [47] as a baseline, the total training loss for the framework is the combination of the supervised loss term and the unsupervised loss term.

In detail, for the labeled data, weak augmentation is applied, and the standard cross-entropy (CE) loss between the prediction logits p_b^l output from the classification network and the ground truths y_b^l is calculated as the supervised classification loss

$$\mathcal{L}_{\mathrm{CLS}}^{s} = \frac{1}{B_l}\sum_{b=1}^{B_l}\mathbb{L}_{\mathrm{CE}}\left(p_b^l, y_b^l\right) \tag{1}$$

where B_l represents the batch size of the labeled training dataset and $\mathbb{L}_{\mathrm{CE}}(\cdot)$ denotes the CE loss function.

As for the unlabeled data, both weak augmentation and strong augmentation operations are simultaneously applied to acquire two different views of the input sample, respectively. We follow the augmentation operations utilized in the classical SSL method except for random or center cropping. To construct the baseline for classification, we adopt the flipping and rotation as weak augmentation operations, and the strong augmentation operations include color transformation, contrast enhancement, and so on. For the unsupervised loss term $\mathcal{L}_{\mathrm{CLS}}^{u_1}$, it regularizes the prediction consistency between weakly augmented view p_b^w and the strongly augmented view p_b^s as

$$\mathcal{L}_{\mathrm{CLS}}^{u_1} = \frac{1}{B_u}\sum_{b=1}^{B_u}\mathbb{I}\left(\max\left(p_b^w\right) \geq \tau\right)\cdot\mathbb{L}_{\mathrm{CE}}\left(p_b^w, p_b^s\right) \tag{2}$$

where B_u represents the batch size of the unlabeled dataset, $\mathbb{I}(\cdot)$ denotes the indicator function, and τ is a confidence threshold for filtering the reliable PLs.

2) Attention-Guided Data Augmentation Strategy: Most of the existing semisupervised classification methods adopt cropping in central or random locations as a typical data augmentation operation for acquiring a new image view. However, different from the natural scene images, for RSIs characterized by randomly and broadly distributed geographic objects [48], the center or random crop operations might falsely crop the target object regions, which lost the object information on the contrary.

To obtain additional self-supervision gain focused on the target regions while guaranteeing the alignment of semantic features before and after augmentation, we devise an attention-guided data augmentation strategy that is simple to insert into the existing consistency-based semisupervised classification framework. This novel view contributes to excavating the image-level semantic clues from the unlabeled remote sensing data and adequately promotes the classification performance of the classifier.

Typically, the CAM technique can help to capture the image object regions that are focused during the training of the classification model. As shown in Fig. 4, the CAM algorithm is introduced to derive the class-aware attention mask $\mathbf{M}_{\mathrm{att}}$ of the existing strongly augmented view $\mathrm{Aug}_s(x_b^u)$ as follows:

$$\mathbf{M}_{\mathrm{att}} = \mathbb{I}\left(\mathcal{F}_{\mathrm{CAM}}\left(\mathrm{Aug}_s\left(x_b^u\right)\right) > \mu\right) \tag{3}$$

where μ denotes the threshold value for binarization, which is set to 0.5 empirically in this article.

Given the binary attention mask $\mathbf{M}_{\mathrm{att}}$ in (3), we further find a bounding box that can cover the whole selected positive

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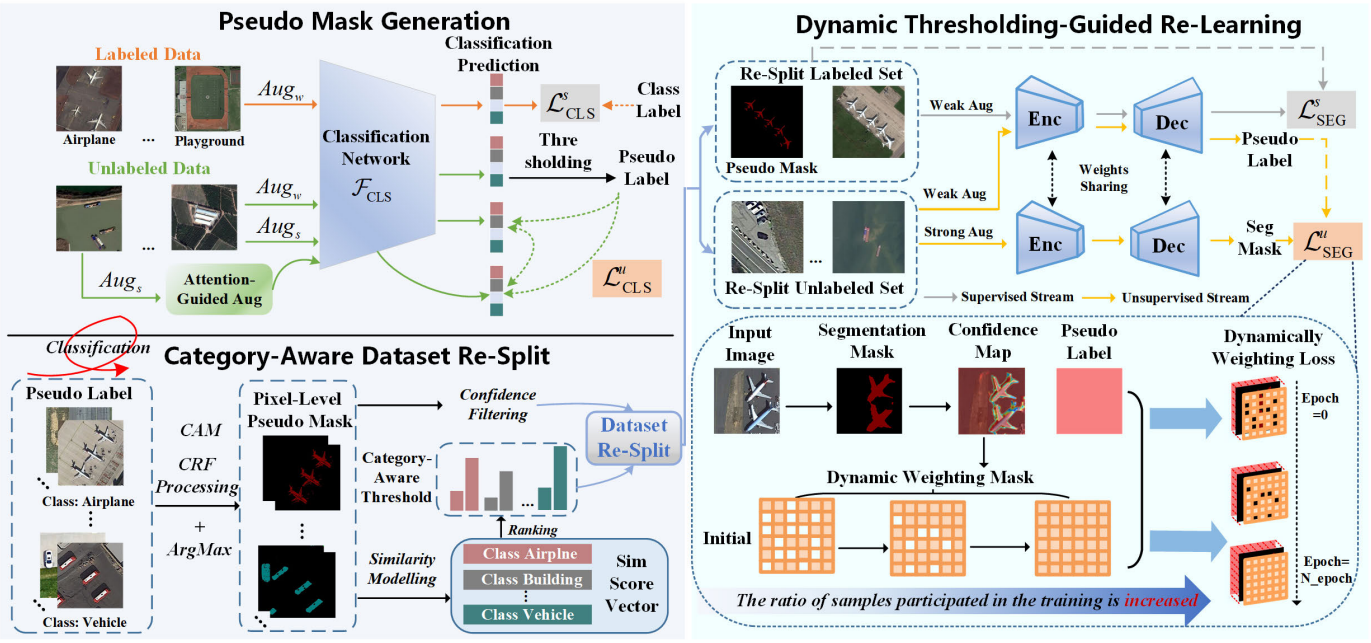


Fig. 3. Illustration of the proposed PSRL framework, which contains three stages. Given the training dataset under extremely weak annotation conditions, we first train a semisupervised classification network to generate pixel-level pseudomasks. Then, to alleviate the adverse effect caused by noisy pseudo supervision, a category-aware dataset resplit strategy is designed to obtain class-balanced reliable and unreliable subsets. Ultimately, a dynamic thresholding-guided semisupervised segmentation network is introduced to relearn from the pseudo supervision.

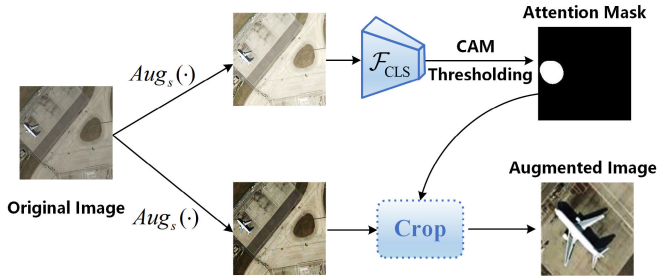


Fig. 4. Illustration of the attention-guided data augmentation strategy.

region of $\mathbf{M}_{\text{att}}$ to crop the original image. In addition, it is then enlarged to the shape of the original input sample as the augmented input data. As illustrated in Fig. 4, by increasing the scale of the object region, the proposed attention-guided data augmentation strategy can help to produce a second strongly augmented sample $\text{Aug}_{s_2}(x_b^u)$, which focuses on the discriminative target regions. The unsupervised loss $\mathcal{L}_{\text{CLS}}^{u_2}$ of the second strongly augmented view can be calculated in the same way as (2).

3) Total Training Loss for Classification Model: Aiming to promote the prediction consistency between the two strongly augmented views, an implicit consistency loss $\mathcal{L}_{\text{CLS}}^{u_{12}}$ between them is involved in the unsupervised loss, which can be defined as follows:

$$\mathcal{L}_{\text{CLS}}^{u_{12}} = \frac{1}{n}\left\|\mathcal{F}_{\text{CLS}}\left(\text{Aug}_s\left(x_b^u\right)\right) - \mathcal{F}_{\text{CLS}}\left(\text{Aug}_{s_2}\left(x_b^u\right)\right)\right\|_2^2 \quad (4)$$

where $\mathcal{F}_{\text{CLS}}(\cdot)$ denotes the classification model and n denotes the total number of vector elements. In this way, the semisupervised network can be enhanced by $\mathcal{L}_{\text{CLS}}^{u_{12}}$ to learn the self-supervised feature representation with invariance and make the network obtain better comprehensive classification performance.

Ultimately, the entire unsupervised term $\mathcal{L}_{\text{CLS}}^{u}$ for the unlabeled data can be formulated as the sum of $\mathcal{L}_{\text{CLS}}^{u_1}$, $\mathcal{L}_{\text{CLS}}^{u_2}$, and $\mathcal{L}_{\text{CLS}}^{u_{12}}$. To sum up, the overall training loss for the classification network is the weighted sum of the supervised loss and the unsupervised loss as

$$\mathcal{L}_{\text{CLS}} = \mathcal{L}_{\text{CLS}}^{s} + \lambda_{\text{CLS}}^{u}\mathcal{L}_{\text{CLS}}^{u} \quad (5)$$

where λ_{CLS}^{u} denotes the balancing weight for the unsupervised classification loss term.

4) Generating Pixel-Level Pseudomask: Given the trained classification model, the pixel-level foreground attention map $\mathbf{M} \in \mathbb{R}^{C\times H\times W}$ is produced by the CAM technique, which combines the final classification feature map using the fully connected layer weights. Then, following [17], the background activation map is estimated as follows:

$$\text{BG}\left(\mathbf{M}_{:,h,w}(x,y)\right) = \left(1 - \max_{1\le c\le C-1}\mathbf{M}_{c,h,w}(x,y)\right)^{\beta} \quad (6)$$

where (x, y) denotes the coordinates of the pixel and β, which is larger than 1, is the decay rate for suppressing the background labels.

To produce more precise pixel-level pseudomasks for segmentation, the multichannel foreground attention map and the background activation map are concatenated followed by denseCRF processing [49] and argmax operation

$$\mathbf{Y}_{\text{pseudo}} = \text{Argmax}\left(\text{CRF}\left(\left[\text{Bg}\left(\mathbf{M}_{:,h,w}\right), \mathbf{M}_{:,h,w}\right]\right)\right). \quad (7)$$

D. Category-Aware Dataset Resplit Strategy

Due to the absence of complete image-level supervision, there inevitably exists a misclassification issue

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in the image-level pseudocategory labels produced from Section III-C, which consequently causes serious image-level noise interferences on the corresponding pixelwise pseudomasks due to semantic category errors. Training deep neural networks in the presence of label noise can be regarded as an issue of learning with noisy labels (LNLs) [50], [51]. Recently, SSL has broadly attracted attention from researchers for LNL in the classification and object detection task [52], [53], [54].

Motivated by the above studies, we transform the issue of noisy PLs into an SSL-based segmentation task to achieve effective relearning from the artificially generated pseudo supervision with uncertainty and noise interference. Specifically, we regard the samples with pseudomasks of higher probabilities to be noisy and unreliable as unlabeled samples; meanwhile, the samples with pseudomasks of higher probabilities to be clean and reliable are regarded as labeled samples. After this reliable–unreliable dataset resplit process, the SSL mechanism can be introduced to facilitate reducing the noisy label interferences as much as possible, meanwhile making full use of the abundant semantic information in the large-scale unlabeled data, which helps to further improve the segmentation precision.

Algorithm 1 Category-Aware Dataset Resplit Algorithm

Input: RSI dataset with pixel-level pseudo masks.
Output: The re-split labeled and unlabeled dataset.
1: Calculate the prediction similarity score ρ of each input sample.
2: For each class c, aggregate the similarity scores of the samples belonging to the class as a similarity vector $\mathcal{R}_c$.
3: Calculate the thresholding values σ^c_{high} and σ^c_{low} according to Eq. (11) and Eq. (12), respectively.
4: **For** each input sample $\mathbf{I}_i$:
5: Calculate the indicator variable g_i according to Eq. (9), and re-split the dataset preliminarily.
6: Calculate the classification confidence score s_i according to Eq. (10), and further refine the initially split dataset.
7: **end**

The above analysis indicates that an appropriate reliable–unreliable pseudomask partition plays a significant role in the performance of pseudo supervision relearning since an extensive amount of incorrect predictions chosen as pseudo-ground truths can lead to suboptimal model performance. The perturbation strategy [55] measuring the prediction variations of perturbations applied on the input is prevalently used to determine the patterns that work in the model decision process, which helps to explain the prediction mechanism of the deep neural network. Inspired by the superior performance of the perturbation strategy in the field of interpretability of deep learning, we propose a category-aware dataset resplit strategy based on masking perturbation, as shown in Fig. 5, which utilizes the prediction variance from the perturbed input to effectively implement the reliable–unreliable pseudo supervision division. The details of the dataset resplit strategy are provided as follows.

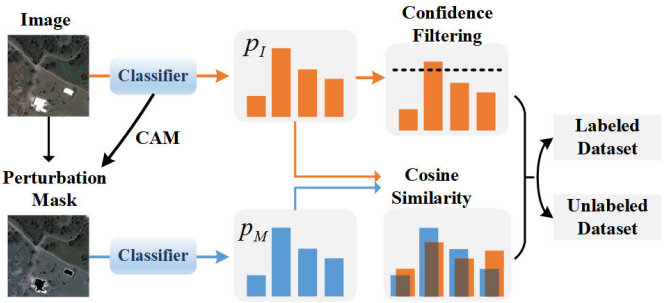


Fig. 5. Illustration of the proposed masking perturbation-based similarity modeling for dataset resplit.

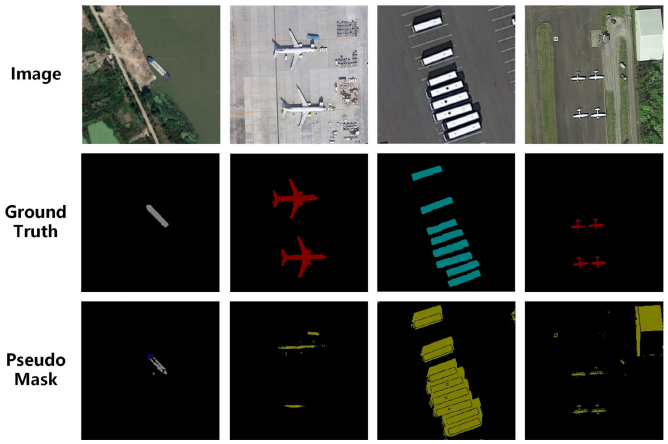


Fig. 6. Illustration of the typical types of noisy pseudomasks generated in the first stage of our proposed PSRL framework. The examples of (a) and (b) indicate that the object regions have been largely missed. As for the examples of (c) and (d), object locations are mainly correct, while the semantic category of the pseudomasks is incorrect.

1) Masking Perturbation-Based Dataset Resplit: We model the image-level similarity score based on the masking perturbation mechanism as the metric for dataset resplit. Formally, given the training samples and the corresponding pseudomasks, select the pixels of nonzero values in the pixel-level pseudomasks, and yield the perturbation mask $\mathbf{M}_P \in \mathbb{R}^{H\times W}$ by setting the value of these pixels to 0 and the other pixels to 1. Then, $\mathbf{M}_P$ is elementwise multiplied with the original sample $\mathbf{I}$ as the masking perturbed sample to eliminate the object regions detected in the pseudomask. Subsequently, the original image and the corresponding masking perturbed image are both input to the trained classification network $\mathcal{F}_{\text{CLS}}$, and embedding similarity score ρ between their output classification prediction logits is calculated as

$$\rho = \text{sim}(\delta(\mathcal{F}_{\text{CLS}}(\mathbf{I} \odot \mathbf{M}_P)), \delta(\mathcal{F}_{\text{CLS}}(\mathbf{I}))) \tag{8}$$

where $\delta(\cdot)$ denotes the softmax activation function and $\text{sim}(\cdot)$ is the cosine similarity measurement function.

As shown in Fig. 6, in the complicated scene of multiclass remote sensing imagery, we observe that there mainly exist two types of detection errors in the pixel-level pseudomasks produced from the trained classification network.

a) Object regions have been largely missed: The image contains the target object of a certain category, while the generated pixel-level pseudomask fails to detect the correct

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object regions. It may merely detect a small part of object regions or almost fail to locate any target regions, as displayed in Fig. 6(a) and (b).

b) Semantic category of the PL is incorrect: As illustrated in Fig. 6(c) and (d), the categories of the generated PLs under extremely weak annotations may be incorrect, e.g., the vehicles and airplanes being detected wrongly as buildings due to their similar color appearances. Therefore, although the detected object regions are assigned with the correct object, their semantic category property is incorrect.

As for the first case, since a large proportion of the target object regions are missed in the detection results, the corresponding masking occlusion will merely affect the image classification predictions slightly, and its similarity score tends to be relatively high. For the second case, its similarity score tends to be comparatively low due to the removal of the target objects.

Therefore, we employ the similarity score as the initial split metric. Pseudolabeled images of similarity score ρ higher than σ_{high} or lower than σ_{low} are both divided into the unlabeled dataset because of their higher probabilities of belonging to the noisy samples. Specifically, an indicator variable is calculated as (9) to denote whether the input sample $\mathbf{I}_i$ has been divided into the unlabeled dataset, in which g_i with the value of 1 means that the input should be divided into the unlabeled subset, while g_i with the value of 0 denotes that the sample belongs to the labeled subset

$$g_i = \mathbb{I}\left(\rho_i \geq \sigma_{\text{high}}\right) + \mathbb{I}(\rho_i \leq \sigma_{\text{low}}) \tag{9}$$

where the two terms of thresholding value are determined by further calculating the ratio of two subsets, which will be introduced in Section III-D2.

However, reliable PLs can also be assigned with lower similarity scores. Therefore, we further introduce a classification confidence score as given in (10) to measure the reliability of the classification results, which is utilized to further filter the noisy pseudomasks

$$s_i = \max(\mathcal{F}_{\text{CLS}}(\mathbf{I}_i)). \tag{10}$$

Specifically, after the preliminary splitting process, we further employ the classification confidence score to refine the splitting results. For samples with g_i equal to 1, if s_i is larger than 0.95, then the input sample $\mathbf{I}_i$ is redivided into the labeled set. For samples with g_i equal to 0 and s_i lower than 0.75, the corresponding input $\mathbf{I}_i$ is refined to the unlabeled set. Here, the threshold values of 0.75 and 0.95 are set empirically.

2) Category-Aware Thresholding: Due to the diverse shapes and varied sizes of geographic objects in the challenging scene of multiclass geographic object segmentation, the learning difficulty of the samples from different categories varies greatly. For example, compared to large-scale geographic objects, small-scale vehicle objects are typically more challenging to locate accurately. Therefore, employing unified high- and low-threshold values for splitting samples with different categories can easily lead to an unbalanced issue, as illustrated in Fig. 7, which can cause severe overfitting on the major classes. Therefore, we take the category balance issue into account and propose a category-aware thresholding strategy. The splitting thresholds and splitting process for each object category are calculated, respectively, to guarantee the balanced proportion of the labeled and unlabeled samples for each category in the redivided dataset.

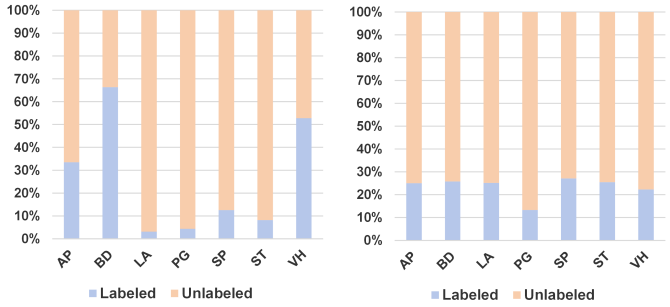


Fig. 7. Illustration of the label-unlabeled split result based on different splitting strategies in seven object classes, i.e., airplane (AP), building (BD), lake (LA), PG, ship (SP), ST, and vehicle (VH). (a) w/o category-aware thresholding-guided split strategy. (b) w/category-aware thresholding-guided split strategy.

Technically, for each object category c, the similarity scores ρ for image samples classified into the corresponding class by the trained classification network $\mathcal{F}_{\text{CLS}}$ are gathered together as a similarity vector $\mathcal{R}_c \in \mathbb{R}^{1\times N_c}$, where N_c denotes the total number of images attributed with the image label of category c. We then sort $\mathcal{R}_c$ in descending and ascending order as $\mathcal{R}_c^{\downarrow}$ and $\mathcal{R}_c^{\uparrow}$, respectively. Finally, according to the predefined reliable data ratio θ, the quantile of the similarity vector is calculated as the upper bound threshold and the lower bound threshold, respectively. The calculation process can be summarized as

$$\sigma_{\text{high}}^{c} = \text{Percentile}\left(\mathcal{R}_c^{\downarrow}, \theta/2\right) \tag{11}$$

$$\sigma_{\text{low}}^{c} = \text{Percentile}\left(\mathcal{R}_c^{\uparrow}, \theta/2\right) \tag{12}$$

where θ represents the splitting proportion of the numbers of the labeled data and the total data, and it is set to 50% according to the experiment analysis. Hence, the Top-$(1-\theta)$ unreliable pseudomasks enter the unlabeled dataset, and the others belong to the labeled dataset.

E. Dynamic Thresholding-Guided Relearning

After dataset resplitting, we construct a semisupervised segmentation network guided by a DTS to further excavate the valuable semantic information from the generated pseudo supervision while alleviating the negative noise interferences caused by incorrect predictions. We first introduce the baseline for the relearning network and then provide the details for the DTS.

1) Baseline Network: We follow the prevalent consistency-based semisupervised segmentation method UniMatch [46] as the baseline for our relearning network $\mathcal{F}_{\text{SEG}}$. The baseline is composed of an encoder and a decoder, which is trained by the combination of a supervised loss and an unsupervised loss as follows:

$$\mathcal{L}_{\text{SEG}} = \mathcal{L}_{\text{SEG}}^{s} + \lambda_{\text{SEG}}^{u} \cdot \mathcal{L}_{\text{SEG}}^{u} \tag{13}$$

where λ_{SEG}^{u} is the weight of the unsupervised loss and $\mathcal{L}_{\text{SEG}}^{s}$ calculates the CE loss between the prediction of labeled

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data and ground truth, while the unsupervised loss term $\mathcal{L}^u_{\text{SEG}}$ regularizes the prediction $\mathbf{M}^{\text{Seg}}_s$ of a strongly perturbed unlabeled image to be the same as that of its corresponding weakly perturbed version $\mathbf{M}^{\text{Seg}}$. In detail, $\mathcal{L}^u_{\text{SEG}}$ can be defined as

$$\mathcal{L}^u_{\text{SEG}} = \frac{1}{B_u}\sum_{b=1}^{B_u} \mathrm{T}\left(x^u_b\right) \cdot \mathbb{L}_{\text{CE}}\left(\mathrm{M}^{\text{Seg}}, \mathrm{M}^{\text{Seg}}_s\right) \tag{14}$$

where $\mathbf{T}$ denotes the PL filtering mask to select the highly reliable PL as the supervision signal for unlabeled sample, which is defined as

$$\mathbf{T} = \mathbb{I}\left(\max\left(\mathcal{F}_{\text{SEG}}\left(\text{Aug}_w\left(x^u_b\right)\right)\right) \geq \gamma\right) \tag{15}$$

where γ denotes the confidence threshold values.

2) Dynamic Thresholding Strategy: However, the fixed threshold employed in (15) is usually a relatively high value to guarantee the quality of PLs, which can cause an issue of insufficient utilization for unconfident yet correct PLs. Aiming to alleviate the negative effect caused by noisy PLs while also improving the utilization of PLs, we design a DTS. By assigning dynamic threshold values, DTS can adaptively filter the more confident predictions from the weakly augmented stream as pseudo supervision for the strong augmented stream in SSL. In this way, the generated PLs can be sufficiently utilized to improve the model performance.

The DTS has been introduced in the field of semisupervised classification to reduce the confirmation bias caused by noisy PLs [56], in which the confidence threshold value is gradually increased to reduce the number of pixels that participated in the training. In this way, relatively more diverse information can be included in the initial training process for faster convergence, while reliable labels are chosen in the later training stage to improve the model performance.

However, we observe that the performance of reducing the number of pixels that participated in the training is inferior to that of increasing the training samples in our task. We consider that the reason might be that diverse PLs in the early training stage can introduce much noise interference for the pixelwise segmentation task. Therefore, as shown in Fig. 8, in contrast to [56], we follow the curriculum learning mechanism to dynamically increase the number of pixels that participated in the training, thus learning from the easier samples to more difficult samples progressively, which is inspired from the mechanism of human learning. It begins training with parts of the most reliable samples to help alleviate model collapse caused by noise interferences in the early training stage and adaptively increases the PLs since the prediction becomes more and more accurate with the network optimization.

To adaptively filter the reliable predictions, we leverage the entropy to estimate the confident degree of a predicted distribution, which is widely used in statistics acting as the degree of disorder or randomness in the system [57]. A lower entropy generally demonstrates a more peaked distribution with less uncertainty and higher confidence. Concretely, the pixelwise confidence map $\mathbf{Conf} \in \mathbb{R}^{H\times W}$ of the segmentation

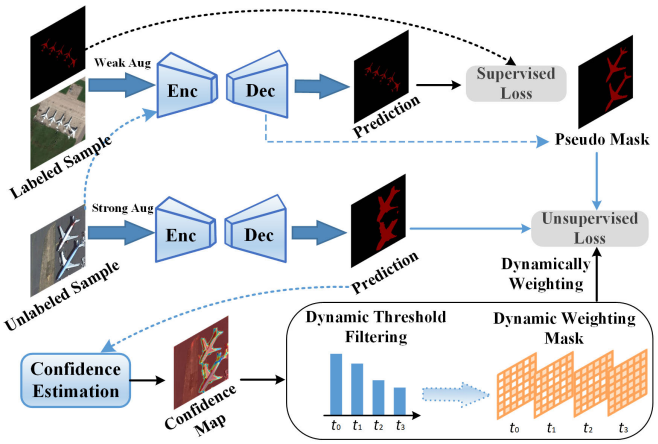


Fig. 8. Illustration of the dynamic thresholding-guided relearning process, in which the unlabeled samples in the resplit dataset are supervised by dynamic weighting loss.

mask can be estimated as follows:

$$\mathbf{Conf}(i,j) = 1 - \left\| -\sum_{c=0}^{C-1} \mathbf{M}^{\text{Seg}}_c(i,j) \log \mathbf{M}^{\text{Seg}}_c(i,j) \right\|_2 \tag{16}$$

where $\mathbf{M}^{\text{Seg}}_c$ denotes the segmentation mask at the cth channel output from the semisupervised segmentation model and (i, j) denotes the pixel coordinates.

Then, we define masks with confidence higher than γ_t as reliable PLs at the training epoch t while the others as unreliable ones that are not qualified for supervision. Thus, a dynamic weighting mask $\mathbf{W} \in \mathbb{R}^{H\times W}$ can be formulated as

$$\mathbf{W}(i,j) = \mathbb{I}(\mathbf{Conf}(i,j) \geq \gamma_t) \tag{17}$$

where the dynamic threshold value γ_t at the tth training epoch is gradually decreased during training, which is set as a quantile corresponding to α_t as follows:

$$\gamma_t = \text{Percentile}(\text{Flatten}(\mathbf{Conf}), \alpha_t) \tag{18}$$

where Flatten$(\cdot)$ represents the operation of flattening a 2-D image into a 1-D vector, $\alpha_t = (100 - \eta) \cdot (1 - t/N_{\text{epoch}})$, N_{epoch} is the total training epochs, and η represents the initial percentage of unlabeled samples participated in the SSL-based training process. In this article, the parameter of η is set to 70, which is discussed in the section of ablation studies.

Therefore, in contrast to conventional unsupervised segmentation loss utilizing a fixed threshold value, we employ the dynamic weighting mask $\mathbf{W}$ to adaptively reweight the unsupervised loss term pixel by pixel. Thus, the dynamic thresholding-guided unsupervised segmentation loss in our relearning process can be calculated as

$$\mathcal{L}^u_{\text{SEG}} = \frac{1}{B_u}\sum_{b=1}^{B_u} W\left(x^u_b\right) \cdot \mathbb{L}_{\text{CE}}\left(M^{\text{Seg}}, M^{\text{Seg}}_s\right). \tag{19}$$

As exhibited in Fig. 9, the confidence maps of input images can reflect the reliability degree of the output pseudomasks to a certain extent, which means that the relatively higher pixel values correspond to more reliable predictions, and vice versa. In addition, the dynamic weighting mask correspondingly

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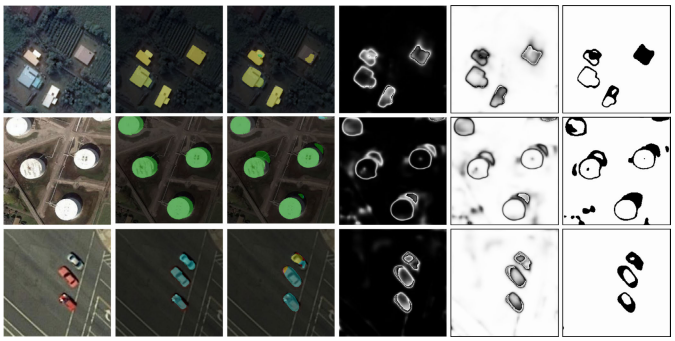


Fig. 9. Visualization of intermediate results of DTS. (a) Training samples. (b) Ground truth. (c) Pseudomask. (d) Entropy map. (e) Confidence map. (f) Dynamic weighting mask.

activates the relatively reliable pixels and inhibits the comparatively uncertain pixels in the pseudomask. In this way, the adverse impact caused by inaccurate prediction in the pseudomask can be alleviated to some extent.

IV. Experiments

A. Experiment Settings

1) Dataset: We evaluate our proposed method on two multiclass remote sensing imagery datasets that are listed below.

1) ORS [58] contains 4199 images, which are divided into 2000/2199 for train/test, respectively. In our experiment, the following seven object classes, i.e., airplane, building, lake, playground (PG), ship, storage tank (ST), and vehicle, are selected as the target geographic object categories to be segmented.
2) EORSSD [59] includes 2000 images, of which 1400 images are for training and the others for testing. In EORSSD, the following ten object classes including airplane, building, car, court, golf course, island, river, road, ship, and water are regarded as the foreground geographic object categories.

2) Implementation Details: In our experiment, only ten samples are labeled with image-level tags for each category in the training set under extremely weak annotations. For the classification model, we apply VGG-16 [60] pretrained on ImageNet [61] as the backbone. To generate pseudomasks, the decay rate β in (6) is set to four following [62]. The learning rate of the two datasets is both 0.001, which is decayed by 1.25 for every 10 epochs. As for object segmentation, DeepLabV3+ [63] with ResNet-101 [64] pretrained on ImageNet is adopted as the backbone. We follow [38] to adopt the poly learning rate decay with a power of 0.9 for training the semisupervised segmentation network. The initial learning rate on the two datasets is 0.02 and 0.04, respectively. Specifically, to better validate the role of unlabeled data of the dataset, we define "Sup Baseline" as the model trained only by the labeled samples in the dataset for further comparison.

3) Evaluation Metrics: To evaluate the performance of multiclass geographic object segmentation approaches, we adopt three quantitative evaluation metrics, including mean intersection over union (mIoU), pixel accuracy (PA), and mean F1 score (mF1). Then, we provide detailed definitions of the above evaluation metrics.

mIoU is a prevalent evaluation metric for the semantic segmentation algorithm, which can comprehensively evaluate pixel-level prediction accuracy, especially when dealing with objects of varied types and sizes. Specifically, mIoU measures the model performance by calculating the ratio of intersection and union between the area of model predictions and the ground truths among all categories, which can be calculated as

$$\text{mIoU} = \frac{1}{C}\sum_{i=1}^{C}\frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \tag{20}$$

where C denotes the total amount of object classes. In addition, true positive (TP) indicates the number of corrected predicted positive samples. False positive (FP) represents the number of samples that are wrongly identified as positive categories. False negative (FN) denotes the number of samples that are wrongly classified as negative categories.

PA is the percentage of the number of correctly predicted samples to the total number of samples, which is given as

$$\text{PA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}. \tag{21}$$

mF1 score is a comprehensive metric that attempts to strike a compromise between recall and precision. The definition of mF1 is given as

$$mF1 = \frac{1}{C}\sum_{i=1}^{C}\frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}. \tag{22}$$

In addition, precision is defined as the ratio of TPs to all the positives predicted by the model, which reflects the ability of the model to correctly recognize positive samples. Recall measures the ratio of all TP samples and the samples that should be predicted as positive samples, which indicates the coverage degree of the model to the positive sample. The two metrics can be calculated as

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{23}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}. \tag{24}$$

4) Comparison Methods: We compare our proposed method with eight SOTA weakly supervised methods under complete image-level supervision. Among the competing models, SEAM [65] utilizes a self-supervised equivalent attention mechanism to impose additional self-supervision constraints on a weakly supervised learning network and narrows the performance gap between the algorithm and the strongly supervised algorithm. By restraining the consistency between the CAM map of the whole image and the CAM map of regional patches, Puzzle-CAM [66] can effectively improve the integrity and quality of CAM results. AMR [67] recalculates the distribution of feature importance by using two complementary branches and accordingly promotes the mining of a more complete target region under the guidance of more accurate weights. ReCAM [68] introduces Softmax CE loss to reactivate the classification model and uses the

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TABLE I
QUANTITATIVE PERFORMANCE COMPARISON ON TWO RSI DATASETS

Method	Supervision	ORS			EORSSD		
		mIoU (%)	PA (%)	mF1 (%)	mIoU (%)	PA (%)	mF1 (%)
SEAM [65]	Weak	36.41	83.89	49.09	30.01	83.98	45.71
Puzzle-CAM [66]	Weak	34.53	80.44	47.58	34.68	93.98	49.92
AMR [67]	Weak	27.25	82.94	38.33	23.53	85.79	36.54
ReCAM [68]	Weak	31.93	75.08	46.65	29.37	77.52	44.83
CPN [69]	Weak	22.20	80.32	47.81	21.47	91.06	52.73
RCA [70]	Weak	39.18	78.31	52.84	**37.61**	88.64	50.12
AFA [71]	Weak	36.23	89.84	69.33	23.32	88.10	57.62
SCD [72]	Weak	36.80	90.89	71.74	25.21	88.36	**61.36**
Sup Baseline	Extreme Weak	20.67	88.09	32.46	16.12	90.46	28.66
Ours-Stage1 w/o AttAug	Extreme Weak	33.46	92.51	46.98	28.89	**96.79**	50.17
Ours-Stage1	Extreme Weak	37.40	90.45	50.82	30.51	95.58	46.55
Ours-Stage1 w/ Seg	Extreme Weak	38.58	93.30	51.54	20.87	96.45	33.64
Ours	Extreme Weak	**43.74**	**93.67**	**57.02**	34.45	95.57	49.87

class reactivation map as a pseudolabel, helping to mitigate the limitations of the initial CAM caused by the adoption of standard CE loss. The CPN [69] method uses images with complementary patch regions to mine more integral foreground activation regions. RCA [70] utilizes region-level semantic contrast and semantic aggregation mechanisms to mine rich semantic context information. As for end-to-end AFA [71] and SCD [72], they both employ the transformer as the backbone encoder to generate more complete PLs. In addition, SCD refines the initial PLs by self-correspondence distillation, while AFA learns the semantic affinity from the transformer to improve the initial PLs.

For fair comparisons, all competing algorithms are trained from scratch on the RSI datasets.

B. Comparison With SOTAs

1) Quantitative Comparison: The quantitative evaluation results on the ORS test set and EORSSD test set are provided in Table I. It can be seen that our proposed PSRL utilizing only ten labeled samples with image tags for each category achieves comparable performance compared with the methods with total image-level annotations. Specifically, PSRL achieves the highest mIoU, PA, and mF1 metric results on the ORS test set, while its performance on the EORSSD test set containing more geographical target categories was slightly lower than that of the methods under weak supervision. We consider the reason might be that the EORSSD dataset is characterized by more diverse geographical target categories and lower interclass separability compared to the ORS dataset, which increases the difficulty of accurately segmenting the targets under the condition of limited supervision.

Overall, compared to the competing methods, our proposed PSRL method can significantly reduce the labeling cost, hence demonstrating more prominent advantages in the aspect of balancing the labeling costs and the model performance.

2) Qualitative Comparison: Figs. 10 and 11 show the visual segmentation results of our proposal and the competing SOTA weakly supervised segmentation methods on the ORS and EORSSD test sets, respectively, which demonstrates that the proposed PSRL framework can achieve high segmentation accuracies for remote sensing scenes with multiscale and multiple geographic objects. We find that under the condition of extremely weak annotations, high-quality segmentation results comparative to the results employing completely weak annotations can be achieved via our proposed framework. The superior performance achieved can be largely credited to the PSRL algorithm's incorporation of category-aware dataset resplit module and dynamic thresholding-guided relearning, enabling it to optimally exploit the rich semantic content inherent in the large amount of unlabeled data within the extremely weak annotation.

3) Comparison With Variant Forms: We compare our proposed three-stage framework with two types of variants with the same baselines and supervision conditions: 1) directly output the CAM results of stage 1, denoted as "*Ours-Stage1*" and 2) follow the typical pipeline of conventional WSOS to use a fully supervised segmentation network for subsequent learning, denoted as "*Ours-Stage1 w/Seg*." The quantitative and qualitative results of the two variants are illustrated in Table I and Fig. 12, respectively.

Compared with "Our-Stage1," the fully supervised segmentation training can improve performance by 1.18% in terms of mIoU on the ORS test set, while it decreases the mIoU by 9.64% on the EORSSD test set. We consider the reason may be that the noise interferences of pseudo supervision in EORSSD are too large to degrade the performance of the fully supervised segmentation model. By contrast, compared with "Our-Stage1," the dataset resplit and PSRL strategy devised in our three-stage relearning framework elevate the mIoU results by 6.34% and 3.94% on the two test sets, respectively.

In addition, compared with the model trained with only supervised loss (Sup Baseline), our mIoU is improved by around 20% on both two datasets, which demonstrates the superiority of the relearning mechanism for making full use of the unlabeled data. From Fig. 12, it can be observed that our proposed PSRL achieves more accurate segmentation results than only using stage 1 and using stage 1 and fully supervised segmentation, showing superior performances, especially in dealing with small objects and background interferences with shadows.

C. Ablation Study

In this section, we discuss the effect of key components in our proposed framework. All experiments are performed

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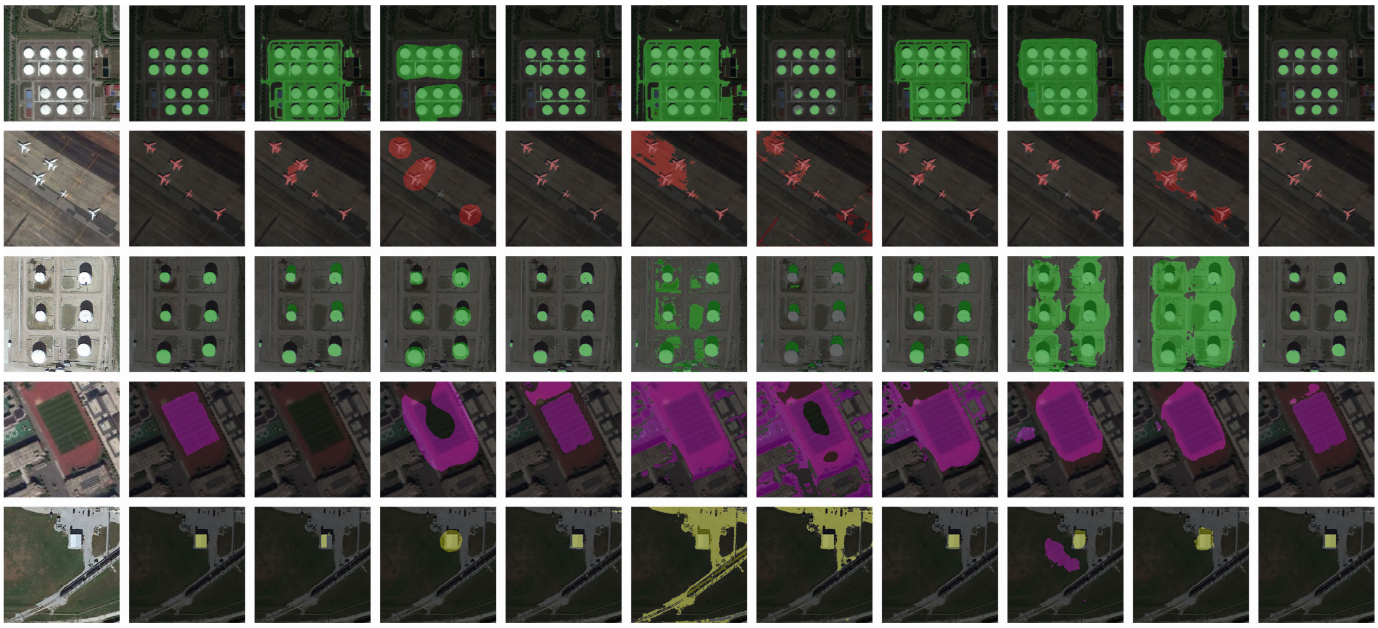


Fig. 10. Comparisons of object segmentation results on the ORS dataset. (a) Original images. (b) Ground truths. (c) SEAM. (d) Puzzle-CAM. (e) AMR. (f) ReCAM. (g) CPN. (h) RCA. (i) AFA. (j) SCD. (k) Ours.

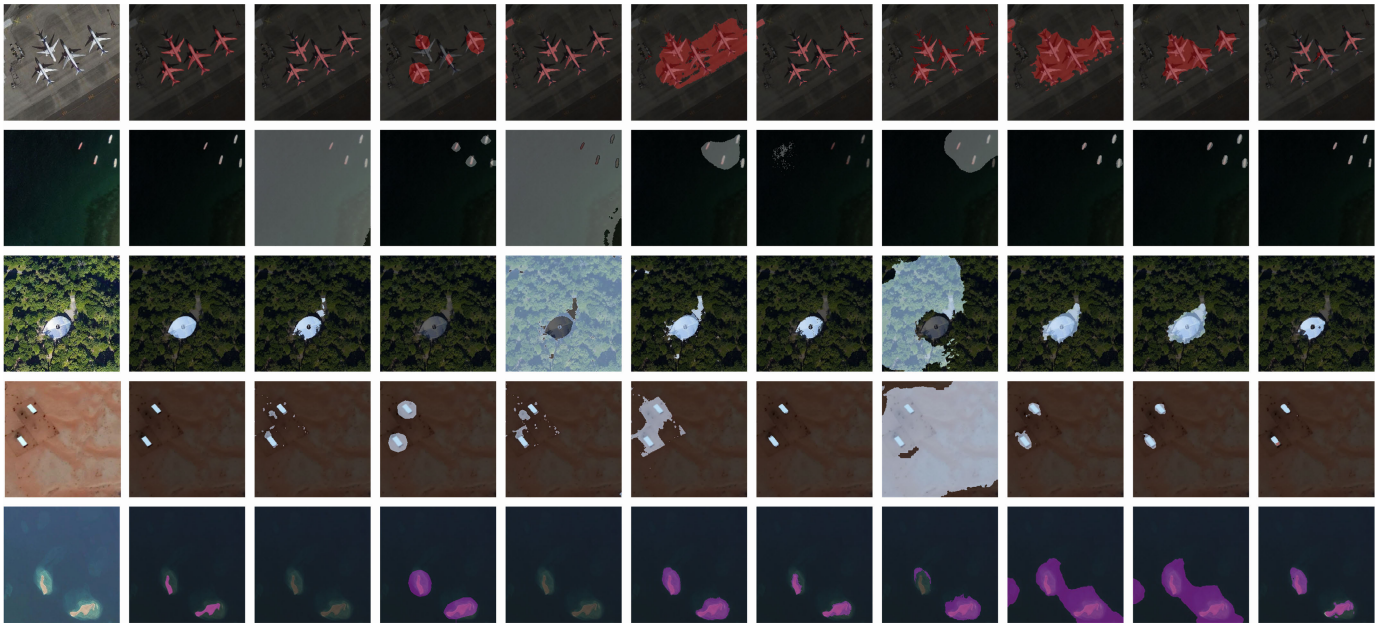


Fig. 11. Comparisons of object segmentation results on the EORSSD dataset. (a) Original images. (b) Ground truths. (c) SEAM. (d) Puzzle-CAM. (e) AMR. (f) ReCAM. (g) CPN. (h) RCA. (i) AFA. (j) SCD. (k) Ours.

on the ORS dataset, and other hyperparameter settings remain the same as the comparative experiments in Section IV-B.

1) Performance of Different Splitting Strategies: We investigate the effectiveness of the category-aware dataset resplit strategy by replacing it with several variants of splitting strategies, including: 1) randomly sampling $\theta\%$ of the generated dataset as a labeled subset, and the rest part is divided as the unlabeled subset, denoted as "*Random Splitting*"; 2) merely using the similarity score as the resplit metric, denoted as "*Similarity Splitting*"; 3) introducing the classification confidence score to further refine the resplit results, denoted as "*Confidence Filtering*"; and 4) employing the category-aware thresholding strategy to calculate the splitting thresholding values, denoted as "*Category Awareness*." The quantitative results of the different variants are reported in Table II. "*Baseline*" in Table II represents the results of stage 1 in the proposed framework.

From Table II, we can find that compared to the baseline, utilizing "*Random Splitting*" for dataset resplit followed by the subsequent relearning stage can improve the mIoU by 2.78%, revealing the necessity of the relearning stage. Furthermore, the step-by-step introduction of "*Similarity*

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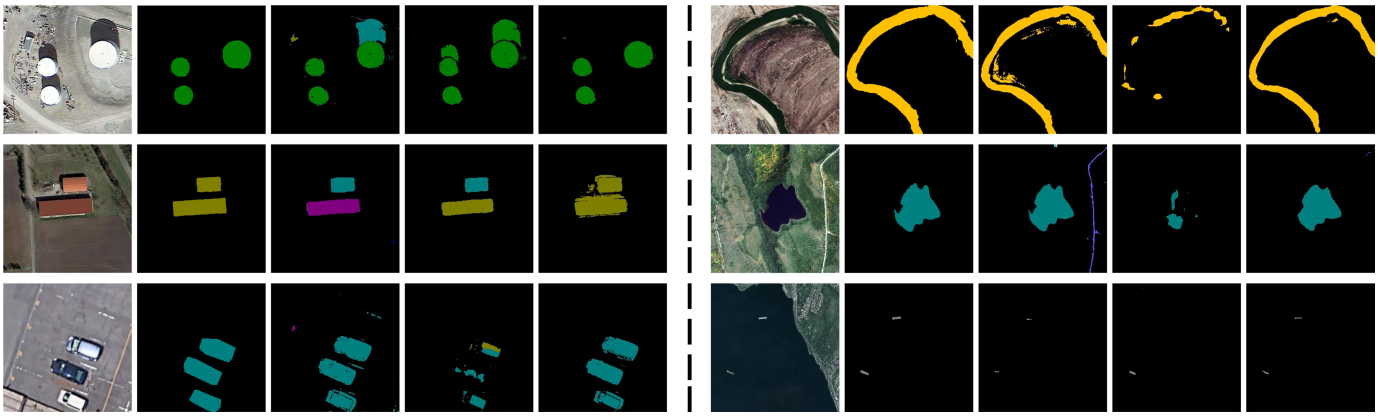


Fig. 12. Qualitative segmentation results on the ORS test (left) and EORSSD test (right) dataset, where the colors represent different semantic categories in the two datasets. From left to right: input images, ground truths, segmentation results of only utilizing Stage 1, Stage 1 w/fully supervised segmentation, and full stages of our proposed framework (PSRL).

TABLE II

MIOU RESULTS OF DIFFERENT VARIANTS OF SPLITTING STRATEGIES

Random Splitting	Similarity Splitting	Confidence Filtering	Category Awareness	Method	mIoU (%)
				Baseline	37.40
✓					40.18
	✓				41.86
	✓	✓			42.05
	✓	✓	✓	Ours	**43.74**$_{(+6.34\%)}$

TABLE III

SEGMENTATION ACCURACY COMPARISON OF DIFFERENT BASELINES AND THRESHOLDING METHODS, WHERE DTS $^{\downarrow}$ DENOTES THE DYNAMIC THRESHOLDING WITH MORE AND MORE SAMPLES

Methods	mIoU (%)	PA (%)	mF1 (%)
FixMatch	31.25	90.23	41.55
FixMatch w/ DTS$^{\downarrow}$	35.74	92.75	47.39
FixMatch w/ DTS	41.50	93.65	54.50
UniMatch	40.97	93.53	54.14
UniMatch w/ DTS$^{\downarrow}$	41.31	93.23	55.08
UniMatch w/ DTS (Ours)	**43.74**	**93.67**	**57.02**

Splitting," "*Confidence Filtering*," and "*Category Awareness*" can progressively elevate the model performance, proving that by category-aware dataset resplitting, our model achieves valid semantic learning from unlabeled data. Meanwhile, the effectiveness of our proposed dataset resplit strategy can be also verified.

2) Performance of Attention-Guided Data Augmentation Strategy: We conduct the experiment of removing the attention-guided data augmentation module from the first stage of our proposed framework, denoted as "*Ours-Stage1 w/o AttAug*," in order to verify the validity of the proposed augmentation strategy.

As shown in Table I, comparing "Ours-Stage1" and "Ours-Stage1 w/o AttAug," we can see that the introduction of our proposed augmentation strategy can lead to an obvious performance improvement. On the ORS dataset, the mIoU and mF1 of "*Ours-Stage1*" are improved by 3.94% and 3.84% against "*Ours-Stage1 w/o AttAug*." Besides, the mIoU metric is increased by 1.62% on the EORSSD dataset. The performance gain demonstrates that the simple attention-guided data augmentation strategy can facilitate the extraction of latent image-level semantic information from the unlabeled remote sensing imagery, thereby boosting the classification performance of the semisupervised classification network. These results also validate the necessity of designing specialized data augmentation modules to obtain new perspectives in the scenarios of RSIs.

3) Performance of Dynamic Thresholding Strategy (DTS): To investigate the influence of DTS on the performance of the final segmentation model, we first compare our proposed algorithm with the method without introducing DTS based on the baseline model of FixMatch and UniMatch, respectively. Hence, in ablation experiments of FixMatch and UniMatch, the confidence threshold is fixed to 0.85 following the setting in FixMatch and UniMatch. Then, to study the effectiveness of gradually increasing the number of training pixels in the DTS strategy, we change the definition of α_t in the DTS strategy as follows:

$$\alpha_t = 100 - (100 - \eta) \cdot \left(1 - t/N_{\text{epoch}}\right). \tag{25}$$

In this way, the thresholding value is gradually increased with the training going on, and the number of pixels in PLs participating in the training process is progressively decreased. Therefore, we denote this experiment as "w/ DTS$^{\downarrow}$." The objective metric results of the above ablation studies are provided in Table III.

First, when employing UniMatch as the baseline model, it can be observed that introducing the DTS module leads to performance gains of 4.78% and 5.61% in terms of the mIoU and mF1 metrics. When utilizing FixMatch as the baseline model, the introduction of DTS obtains gains of 8.46% and 10.59% in terms of the mIoU and mF1. The above results demonstrate the validity of DTS in different types of baseline models. Considering that employing UniMatch as a baseline model can lead to better performances, we use UniMatch as the baseline in the stage of PSRL in our proposed framework.

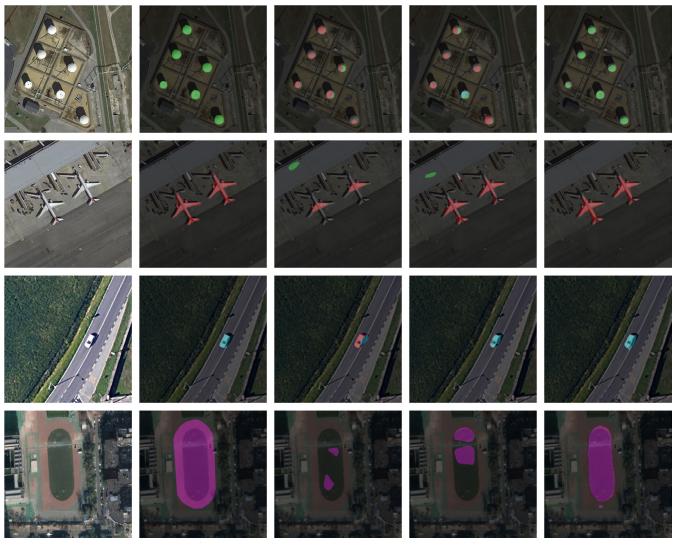
Furthermore, the mIoU of "w/ DTS$^{\downarrow}$" is decreased by 2.43% and 3.97% compared to that of "w/ DTS" with the baseline of UniMatch and FixMatch, respectively, which proves the necessity of increasing the trained pixels' ratio in DTS.

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TABLE IV

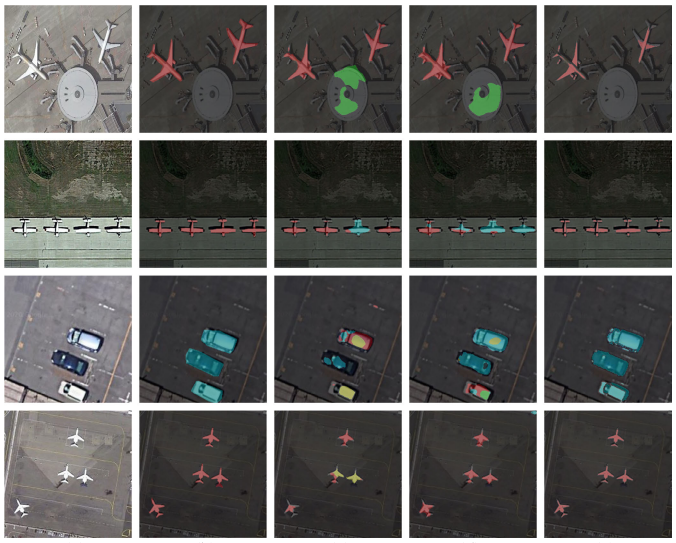
QUANTITATIVE RESULTS UNDER DIFFERENT SPLIT PARTITIONS IN OUR SETTING OF EXTREMELY WEAK ANNOTATIONS, I.E., $n\%$ AS LABELED DATA AND $1-n\%$ AS UNLABELED DATA IN THE TRAINING DATASET

Method	ORS Dataset				EORSSD Dataset		
	Labeled Proportion	mIoU (%)	PA (%)	mF1 (%)	mIoU (%)	PA (%)	mF1 (%)
Sup Baseline	1.75%	18.52	90.43	30.58	14.74	87.33	25.45
	3.5%	20.67	88.09	32.46	16.12	90.46	28.66
	10.5%	22.03	91.15	33.72	16.15	94.60	29.58
	31.5%	23.94	90.21	34.10	19.53	96.24	33.02
Ours	1.75%	36.59	93.23	51.19	31.58	94.87	44.53
	3.5%	43.74	93.67	57.02	34.45	95.57	49.87
	10.5%	44.83	94.87	58.06	35.19	95.98	51.72
	31.5%	46.28	93.75	60.18	35.81	96.74	52.60



(a) (b) (c) (d) (e)

Fig. 13. Visual comparison for the ablation study on the ORS dataset. (a) RSIs. (b) Ground truths. Results are given by (c) FixMatch baseline, (d) FixMatch w/ DTS$^{\downarrow}$, and (e) FixMatch w/ DTS.



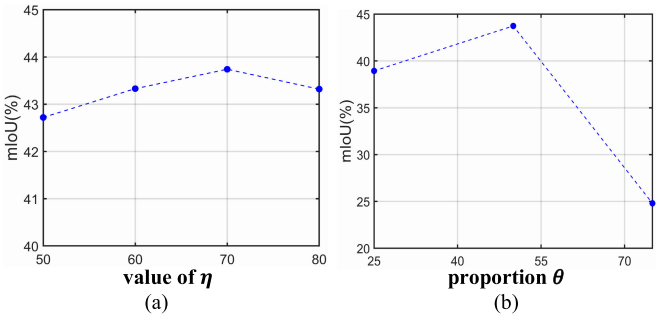
(a) (b) (c) (d) (e)

Fig. 14. Visual comparison for the ablation study on the EORSSD dataset. (a) RSIs. (b) Ground truths. Results are given by (c) UniMatch baseline, (d) UniMatch w/ DTS$^{\downarrow}$, and (e) UniMatch w/ DTS (Ours).

As illustrated in Figs. 13 and 14, compared with the baseline of FixMatch and UniMatch, the introduction of DTS and DTS$^{\downarrow}$ can greatly enhance the detection accuracy in improving the accuracy of the semantic category and eliminating the background interferences.

Finally, we study the effect of different values of η applied in DTS on the model performance. From Fig. 15(a), it can be observed that adding a small change of η does not have any significant effect on the result of mIoU, and the optimal value of η is 70. Therefore, we set the value of η to 70 in the experiment section.

4) Performance of Different Splitting Proportions: To investigate the influence of different splitting proportions θ of the stage of dataset resplit, we conduct a series of ablation studies by varying the value of θ from 25% and 50% to 75% on the ORS dataset, respectively. Fig. 15(b) shows the mIoU results of utilizing different values of θ. It can be observed obviously that $\theta = 50\%$ performs better than both the low value of 25% and the high value of 75%. The reason may be that the splitting proportion θ of labeled and unlabeled subsets determines the reliability of PLs in the resplit dataset. Relatively small θ can lead to too limited supervision for model convergence, while large θ will bring an excess of noise interference from low-quality pseudomasks. Hence, we adopt $\theta = 50\%$ in our experiment.



(a) (b)

Fig. 15. Miou results of utilizing (a) different values of η in DTS and (b) different proportions θ in the stage of dataset resplit.

5) Performance of Different Proportions of Labeled Samples in Extremely Weak Annotations: In Table IV, we explore the influence of the proportion of labeled samples in the training dataset and provide the evaluation results of the segmentation output under different partitions.

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TABLE V
COMPARISON OF MIOU (%) BY UTILIZING DIFFERENT WEIGHTING COEFFICIENT FUNCTIONS FOR DYNAMICAL WEIGHTING

Weighting Function	Linear	Cosine	Log	Sigmoid
mIoU (%)	**43.74**	41.25	42.86	42.15

TABLE VI
PERFORMANCE COMPARISON OF PSEUDOMASKS FROM STAGE 1 WITH DIFFERENT COEFFICIENTS λ^u_{CLS} OF UNSUPERVISED CLASSIFICATION LOSS

λ^u_{CLS}	0.5	1	1.5	2	2.5
mIoU (%)	34.05	**37.40**	32.59	36.43	35.64
PA (%)	93.09	90.45	92.82	**93.17**	92.97
mF1 (%)	47.34	**50.82**	45.36	50.54	49.60

When we change the labeled partition for the ORS dataset under 1.75%, 3.5%, 10.5%, and 31.5%, the segmentation results of our method achieve the mIoU of 36.59%, 43.74%, 44.83%, and 46.28%, respectively. The similarly increasing trend of the quantitative metrics also appears on the EORSSD dataset. The above phenomenon indicates that with the increase in the proportion of labeled samples in the training set under extremely weak annotations, the performance of the model is consistently improved, indicating that leveraging an increased number of labeled samples exerts a favorable influence.

In addition, it can be observed that under different labeled proportions, our model attains a steady improvement compared with the results of the model only trained with the supervised loss (denoted as "Sup Baseline"). The above phenomenon reveals the superiority of our proposed model in adequately excavating and exploiting the semantic information contained in the unlabeled samples, which effectively assists the learning of the supervised stream.

6) Performance of Different Weighting Functions for Dynamical Weighting: Table V reports the results of applying four types of weighting functions in DTS, including the linear function, the cosine function, the logarithmic function, and the sigmoid function. We can find that the linear weighting function obtains the best value of 43.74% in terms of mIoU. The reason may be that the linear function can stably adjust the ratios of samples activated for training compared with other functions.

7) Parameter Analysis of the Unlabeled Classification and Segmentation Loss Weights: In this part, we conduct the parameter analysis of the weighting coefficient λ^u_{CLS} and λ^u_{SEG} of the unlabeled classification and segmentation loss, respectively. In detail, we assign the value of λ^u_{CLS} from 0.5 to 2.5 and the value of λ^u_{SEG} from 1 to 3.5, respectively.

As shown in Table VI, the mIoU and mF1 metrics of the generated pseudomask achieve the highest value when λ^u_{CLS} is set to 1, which means that setting the same importance on the supervised classification loss term and the unsupervised one can greatly boost the model performance. Therefore, in the experiment section, the value of λ^u_{CLS} is set to 1. In addition, as illustrated in Table VII, we can observe that the quantitative metrics of the segmentation results achieve the highest values when λ^u_{SEG} is set to 3. Thus, we set the value of λ^u_{SEG} to 3 in our experiment.

TABLE VII
PERFORMANCE COMPARISON OF SEGMENTATION RESULTS WITH DIFFERENT COEFFICIENTS λ^u_{SEG} OF UNSUPERVISED SEGMENTATION LOSS

λ^u_{SEG}	1	1.5	2	2.5	3	3.5
mIoU (%)	40.81	41.52	41.60	42.88	**43.74**	39.63
PA (%)	93.39	93.52	93.50	93.64	**93.67**	93.27
mF1 (%)	54.00	54.50	55.13	56.00	**57.02**	52.95

V. CONCLUSION

In this article, we present a novel task condition for object segmentation, namely, extremely weak annotations, in which the annotation accuracy and quantity are both limited and facilitate further reducing the labeling cost compared to the weak supervision in remote sensing imagery. To excavate abundant semantic information from the massive unlabeled training data, we propose a novel three-stage learning framework for remote sensing object segmentation under extremely weak annotations, namely PSRL. First, an attention augmentation-guided semisupervised classification model is trained to achieve image-level semantic learning to generate the initial pseudo supervision. Then, we design a category-aware dataset resplit strategy based on the masking perturbation mechanism to filter the reliable high-quality pseudo supervision and resplit the initial dataset, which helps to alleviate the adverse effect caused by the lack of abundant true category labels. Finally, a dynamic thresholding-guided relearning network is trained by the resplit dataset, aiming at further exploring the inherent semantic clues in pseudo supervision while also reducing the confidence bias introduced by PLs. Experiments on two public RSI datasets revealed that our method outperforms baselines under the same supervision condition. It can also achieve comparable performance compared with the methods with complete image-level annotations, which demonstrates that the proposed method can better trade off the labeling cost and the model performance.

However, there are still some limitations in our proposed method. Considering the relatively less supervision information included in the extremely weak annotations, we mainly designed the PSRL framework for the remote sensing scene with single-class geographic objects selected as the target object category to be segmented. Meanwhile, the effect of image degradation on object segmentation is not taken into consideration. For future works, we will further explore the extremely weak annotated semantic segmentation in more complicated and practical remote sensing scenes. In addition, studying to improve the performance under haze and cloud interference could also be planned.

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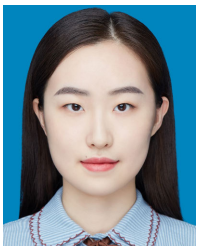
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